



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

Differential and Integral Calculus 1 M - 104041

Exam B

February 21st, 2024

Your ID Number:

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First name and Surname:

Guidelines

1. **Duration: 3 hours.** The use of personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
2. Please, write your name and your ID number on the cover page.
3. The answers to the exam should be written in blue/black pen only. DO NOT use a pencil or correction fluid. DO NOT write near the staple area, otherwise it will be cut off.
4. Explain all your solutions and quote all the theorems you use. **No credits will be given for non-justified answers!** Write clear and complete answers for each problem on the same page.

1. (15p) Consider the following function $f(x)$, with a and b real numbers.

$$f(x) = \begin{cases} \frac{(x-1)^2 - a}{(x-1)^2 + 1} & \text{if } x \leq 1, \\ -x + b \ln(x) + \frac{(x-1)^2}{4} - 3, & \text{if } x > 1. \end{cases}$$

- (i) Find all the possible values for a and b such that $f(x)$ becomes continuous and differentiable on $\mathbb{R}$.

Considering the values, a and b , that have been found in the point (i)

- (ii) Determine **all** the local and global extreme values of the function on the interval $[-1, 3]$, and indicate if they are local maximum or minimum values.
- (iii) Determine the global extreme values of the function on its domain or explain why the function does not have global maximum or minimum values.

Answer

2. (20p) Consider the function

$$f(x) = \begin{cases} \frac{\sin(x)}{5 + 5 \cos(x)} & \text{if } -\pi < x \leq \frac{\pi}{2} \\ -\frac{x^2}{x^2 + \pi^2}, & \text{if } \frac{\pi}{2} < x. \end{cases}$$

- (a) Find all the vertical and horizontal asymptotes.
- (b) Determine $f'(x)$, the domain of $f'(x)$ and the intervals where the function $f(x)$ increases and decreases.
- (c) Find the inflection points and indicate the intervals where the function is concave upward and concave downward.
- (d) Sketch the graph of the function.

Answer

3. (20p) Evaluate the following integrals

(a) $\int (\cos(3x) - x^2) (\sin(3x) - x^3)^5 dx$

(b) $\int_1^9 \frac{dx}{x(2\sqrt{x} - 1)}.$

(c) $\int \frac{\sqrt{9 - x^2}}{x^2} dx$

(d) $\int_1^\infty \frac{\ln(x)}{x^7} dx$

Answer

4. (10p) Find the length of the curve $y(x) = \int_1^{x^2} \sqrt{7t^{1/4} - \frac{1}{4t}} dt$, for $1 \leq x \leq 2$.

Answer

5. (20p) (a) Determine the radius of convergence and the interval of convergence of the following power series.

$$\sum_{n=1}^{\infty} \frac{\pi^n x^n}{n^\pi}$$

- (b) Find the Maclaurin series, i.e. the Taylor series centered at zero, for $\ln(1+x)$ and hence that for $\ln\left(\frac{1+x}{1-x}\right)$. *Hint:* use the fact that $\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$.

Answer

6. (15p) For each of the following statements answer TRUE or FALSE and justify your answer.

(a) There exist at least a real value x such that $x^3 + 1 = -2x + \frac{1}{2}$.

(b) $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^{x/2} = 1$.

(c) If g is a function such that $\int_{1-x^2}^1 g(t)dt = 2x^2 \cos(x)$, then

$$g(t) = -2(1-t) \sin(1-t) + 2 \cos(1-t).$$

Answer